

	Work Instruction	Doc. Revision	01
	AXI V810 Rail Width Motor Kit Removal and Replacement Procedure	Effective Date	20 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	22 Dec 2015	First Release	CHIN CHEE YAN
01	20 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

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Rail width Motor Kit Removal

1. Turn off Air Pressure Regulator.
2. Power off the SSCA main power # Important
3. Open front and rear access door for ease of outer barrier removal.
4. If left AWA rail motor kits need to change, disconnect all cables (Controller bulb head, Pneumatic input valve, Interlock Switch 1and 2) as shown at Left Outer Barrier. **If “Right AWA rail kit” need to replace, remove cables at “Right Outer Barrier”.**

Remark, label marking the Interlock Switch 1 and 2 connector before remove.

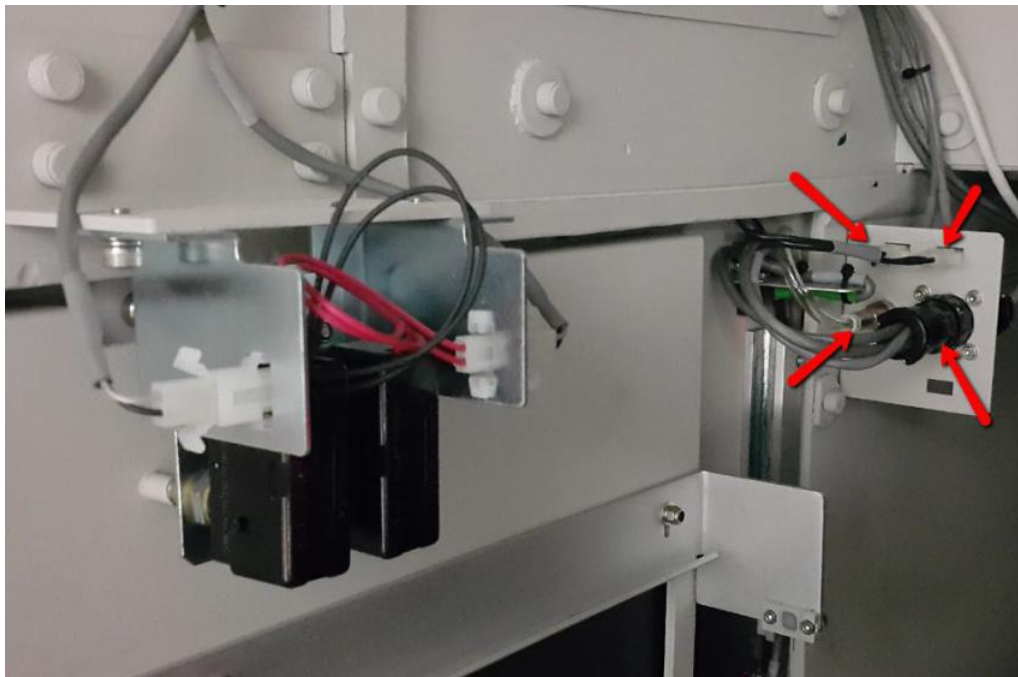


Figure 1 Bulk Head, Interlock Connector & Tubing Removal

5. By using tamper proof tools, remove 4 screws & remove the Outer Barrier door.



Figure 2 Tamper Proof Screws Removal (V810 Std)

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a. For S2EX and XXL machine,

Remove 2 screws as below Figure 3.



Figure 3 Tamper Proof Screws Removal (V810 XXL/S22EX)

6. Move the stage near to outer barrier for ease of parts removal.

7. Check all screws are tight as below Figure 4.

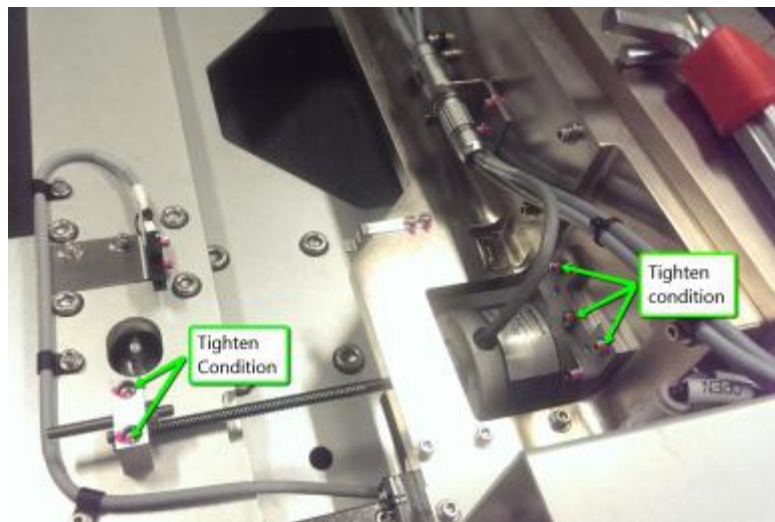


Figure 4 Screws Fastening Condition

8. Disconnect the AWA motor connector from V810 Cable, Motor Stepper AWA Axis as below Figure 5.



Figure 5 Stepper Motor Cable Disconnect

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9. Unscrew the end of AWA lead screw (6-32 x 7/8LG, SHCS, SS screw) from BRACKET, AWA LEAD SCREW (P/N, 4XAXY-070) as shown. **Caution :Inch Size Screw and washer require to reuse.**

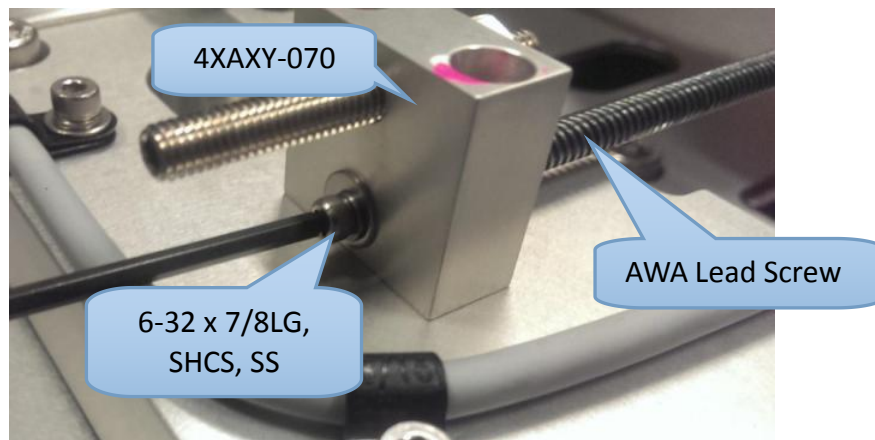


Figure 6 Lead Screw Removal

10. Unscrew (6-32 x 1/2LG, SHCS, SS) another end for AWA lead screw from BRACKET, AWA LEAD SCREW (P/N, 4XAXY-074) as shown. **Caution :Inch Size Screw and washer require to reuse.**

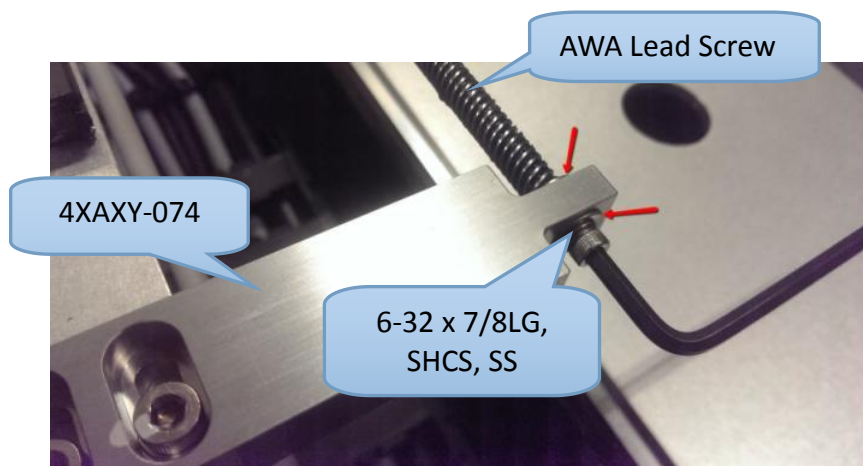


Figure 7 Lead Screw Removal

11. Loosen 2 screws at 4XAXY-074.

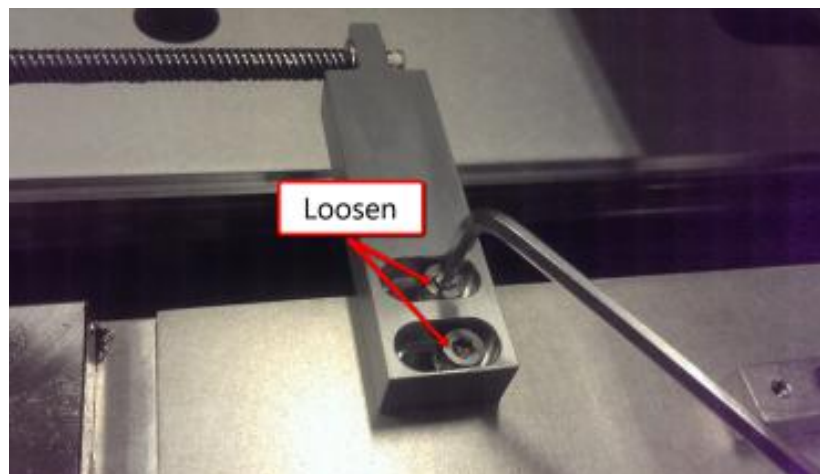


Figure 8

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12. Slowly spin the AWA lead screw with counter clock wise direction and remove from the Rail width motor kit.



Figure 9 Lead Screw Removal (V810 Std)

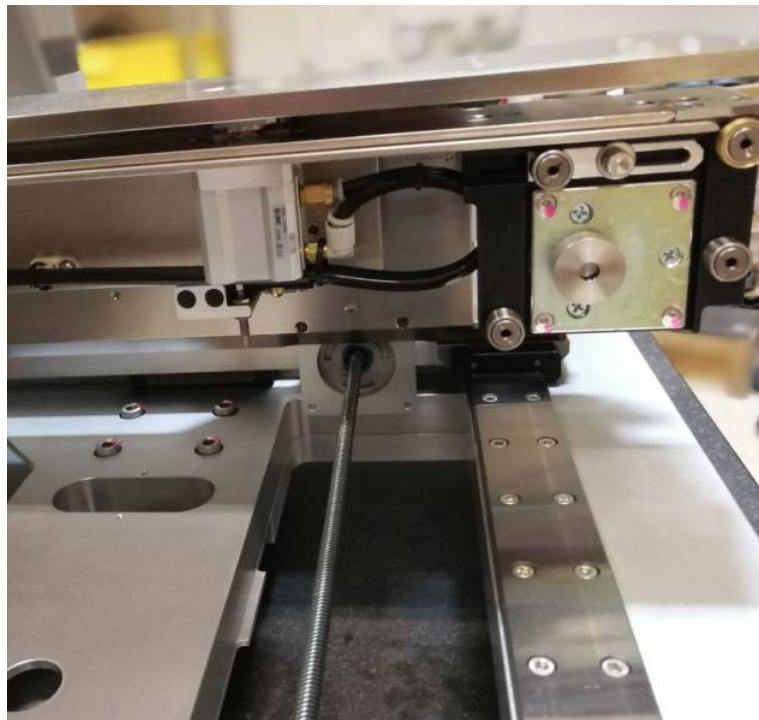


Figure 10 Lead Screw Removal (V810 XXL/S2EX)

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13. **Marking the origin level position** and unscrew (M6x16mm SHCS screws + M6 lock washer) as shown and remove AWA motor with Motor Mount Bracket (P/N, 4XAXY-073) from SUPPORT, GUIDE MOVING (P/N, 4XAXY-113)

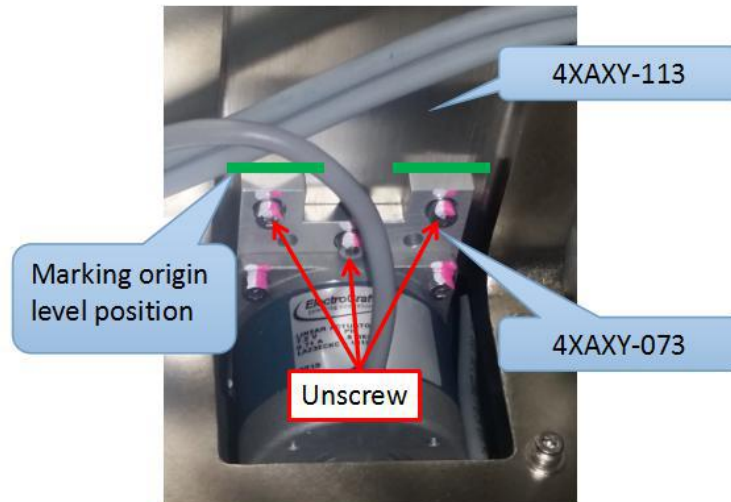


Figure 11 Stepper Motor Bracket Removal

14. Unscrew 4 units of M4x12mm SHCS, SS screws + M4 lock washer as shown and completely remove AWA motor from 4XAXY-073 (Motor Mount Bracket).



Figure 12 Bracket Removal

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Rail width Motor Kit Replacement.

15. Prepare new part for V810 Rail width motor Kit (By refer to below table) as shown. # Incoming part with motor and lead screw.

Model	Part Number	Part Description
V810 Std S1/S2	4XASPP-A030	AXI V810 STD/S2EX AWA Motor Kit
V810 XXL S1/S2	4XASPP-A023	AXI V810 XXL AWA Motor Kit
V810 S2EX	4XASPP-A030	AXI V810 STD/S2EX AWA Motor Kit

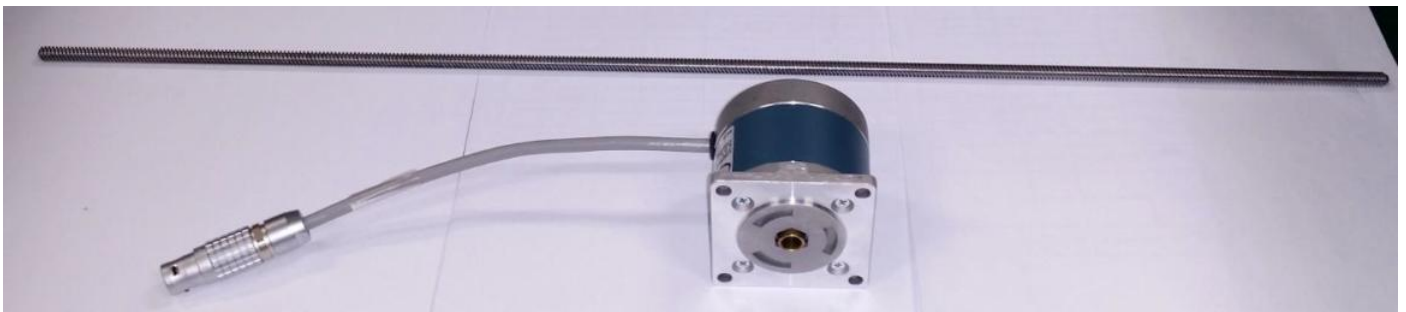


Figure 13 V810 Rail Width Motor Kit

16. Assembly the AWA motor with 4XAXY-073 screw and tighten 4 units of M4x12mm SHCS, SS screws + M4 lock washer as shown. Ensure the cable of motor is facing top side.

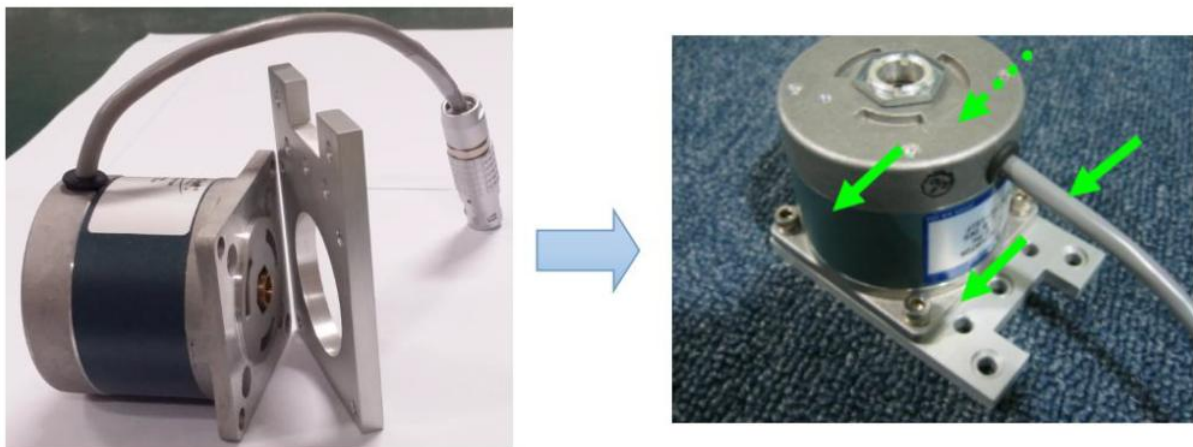


Figure 14 AWA Motor Bracket Assemble

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17. Use L ruler and perform perpendicular alignment between Motor & Motor Bracket (4XAXY-073)

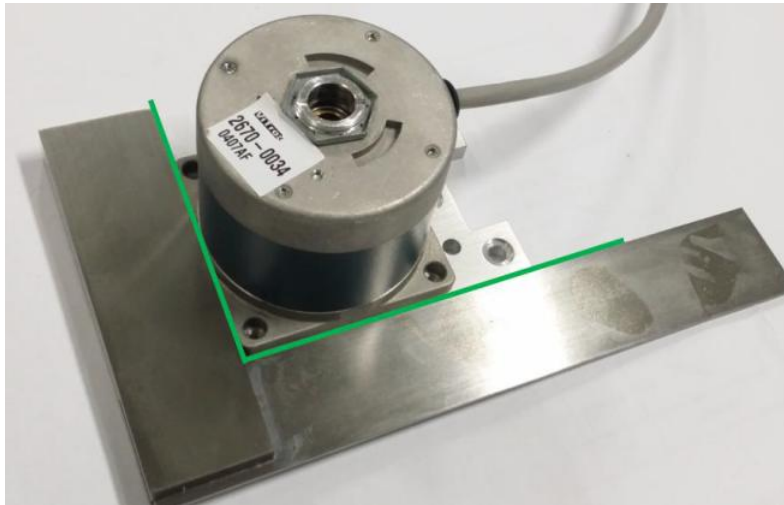


Figure 15 Perpendicular Alignment

18. Assembly the motor and motor mount back to 4XAXY-113 with take the marking as reference.

Tighten all the screws (M6x16mm SHCS screws + M6 lock washer) as shown.

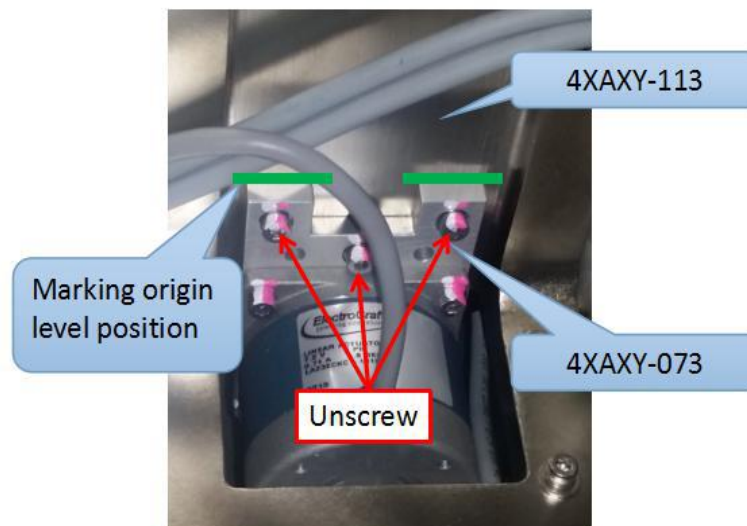


Figure 16 AWA Motor Assembly (Adj Rail)

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19. Slowly spin and insert the new lead screw through the motor until the end is collide with 4XAXY-070.

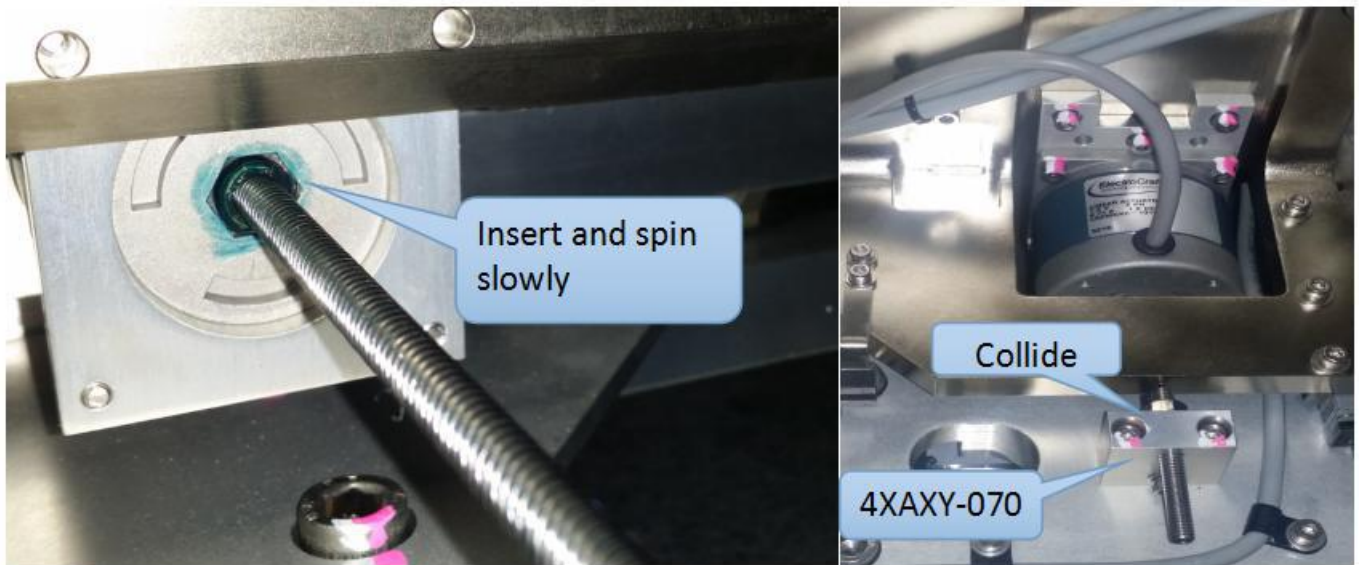


Figure 17 Lead Screw Assembly

20. Screw the end of AWA lead screw with 6-32 x 7/8LG, SHCS, SS screw with 4XAXY-070 as shown. Caution, ensure the spring washer and washer attached, and tighten the screw.

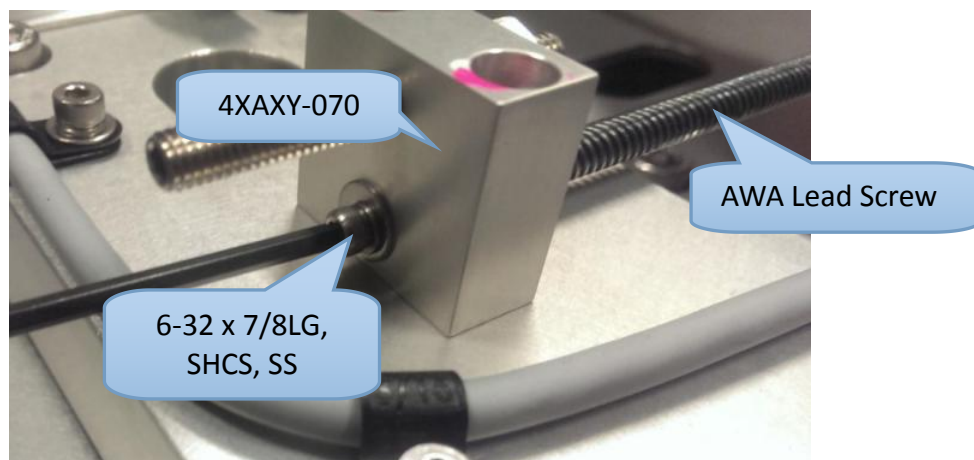


Figure 18 Lead Screw (End) Fastening

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21. Screw another end for AWA lead screw with 6-32 x 1/2LG, SHCS, SS screw with 4XAXY-074 as shown.

Caution, ensure the washer is applying for inner and outer side as shown, and tighten the screw.

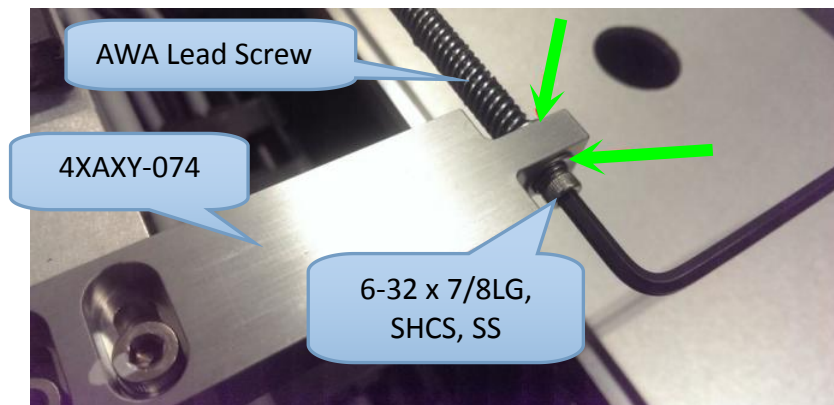


Figure 19 Lead Screws (Front) Fastening

22. Place the L ruler as shown, ensure the 4XAXY-047 is align perpendicular and tighten the screws as shown.

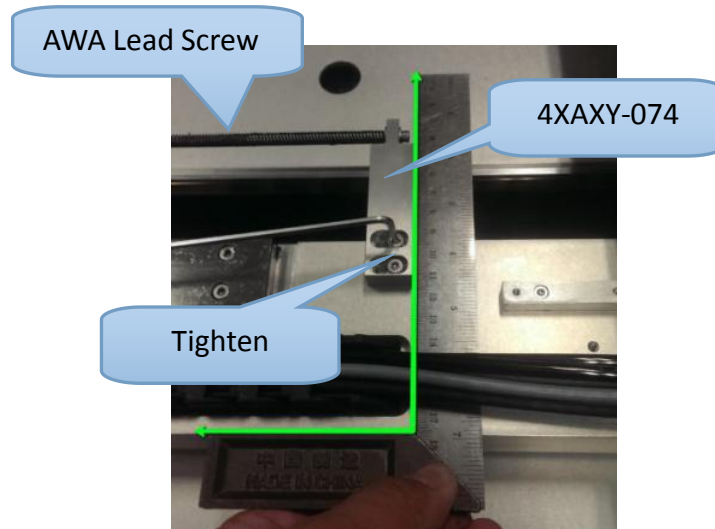


Figure 20

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23. Visual check on the lead screw condition, ensure the lead screw is perpendicular assembled (**No Bend**). If not, repeat step 19~22 and re-align again.



Figure 21 Lead Screw Visual Check

24. Disconnect the AWA motor connector from V810 Cable, Motor Stepper AWA Axis as shown.

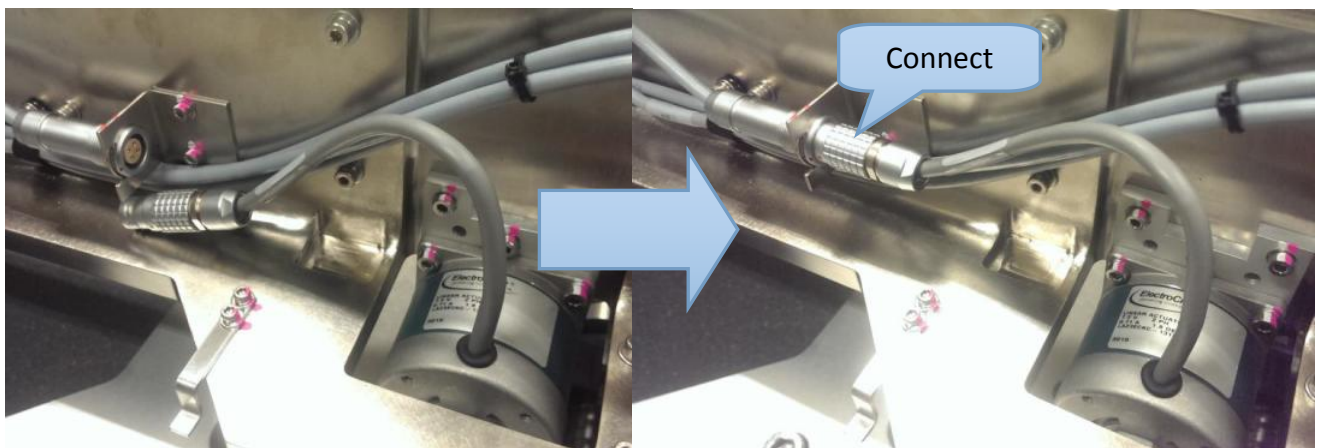


Figure 22 AWA Motor Connection

25. Remove all tools from XY stage.
26. Slightly apply small portion of lubricant grease on the new change AWA motor lead screw & remove excessive grease as below Figure.

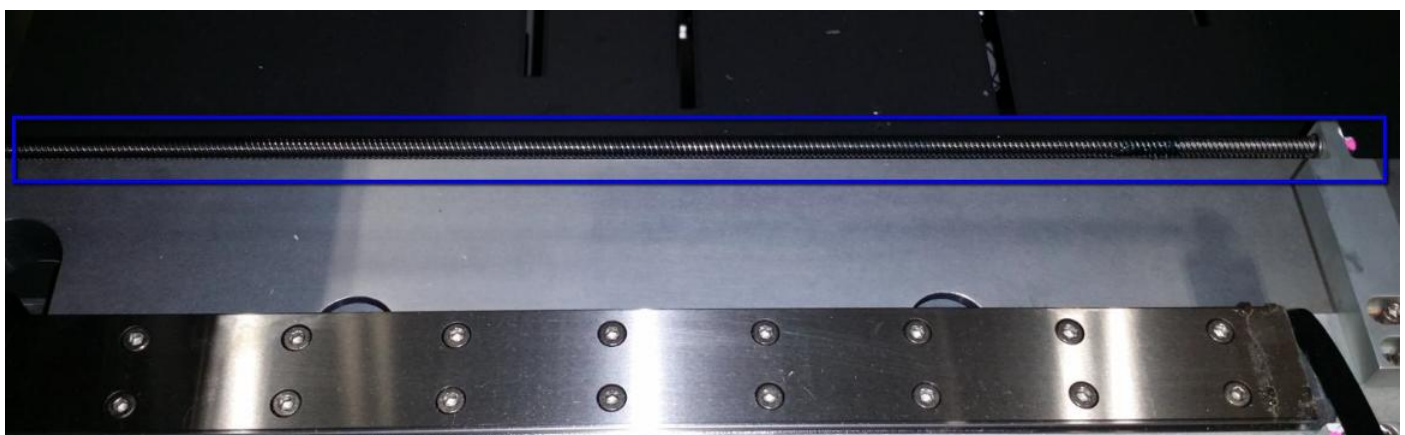


Figure 23 Lead Screw Lubrication

27. Check all screws are fully tighten.

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28. Reassemble the left outer barrier back to origin and reconnect all the cables connection.

29. Close all door access and power on machine and start-up machine as usual.

30. Go to Panel Handling tab, change the display unit to “MM” and click “Initialize”.

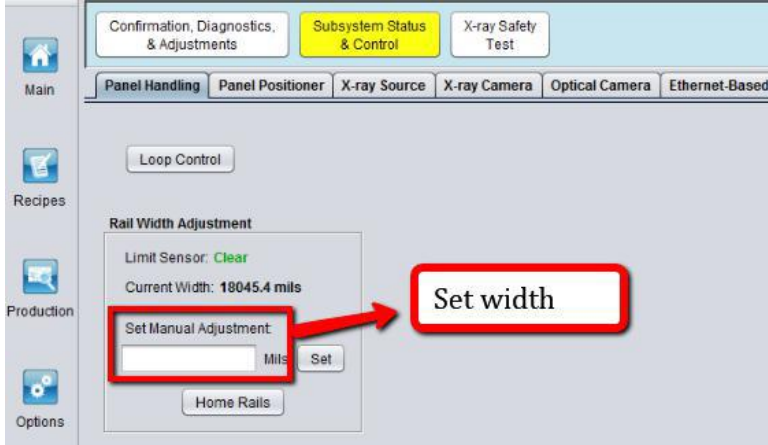


Figure 24 Panel Handling Interface

31. Manual set the rail with 150mm/250mm and 350mm. Measure the actual rail width with Caliper.

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32. Check limit as below table.

Item	Parallelism Limit
(Right)Width Measurement Deduct (-) (Left)Width Measurement	+/-0.125mm
Item	Width Accuracy Limit
Set Width (May refer to below Figure) Deduct (-) Actual Width Measurement Average *[Left Width Measurement + Right Width Measurement]/2 *	+/-0.125mm
	

33. If limit is fail as above table, please refer to below model to perform Rail Width Adjustment

Model	Procedure
V810 Std S1/S2	AXI STD Rail Width Adjustment and Verification
V810 XXL S1/S2	AXI XXL Rail Width Adjustment and Verification
V810 S2EX	AXI S2EX Rail Width Adjustment and Verification